

Math 526, S06
Quiz 7

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $\mathbf{x} = \begin{bmatrix} 1 \\ -12 \\ 3 \\ 4 \end{bmatrix}$ and $\mathbf{A} = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & 3 & -4 & 1 \\ 2 & -5 & -2 & 1 \\ -1 & -1 & -4 & 2 \end{bmatrix}$.

a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$.

Solution:

$$\|\mathbf{x}\|_1 = 1 + |-12| + 3 + 4 = 20$$

$$\|\mathbf{x}\|_2 = \sqrt{1^2 + (-12)^2 + 3^2 + 4^2} = \sqrt{170}$$

$$\|\mathbf{x}\|_\infty = \max\{1, -12, 3, 4\} = 12$$

b) Compute $\|\mathbf{A}\|_1$ and $\|\mathbf{A}\|_\infty$.

Solution:

$$\|\mathbf{A}\|_1 = \max\{1 + 2 + 2 + 1, 2 + 3 + 5 + 1, 3 + 4 + 2 + 4, 1 + 1 + 1 + 2\} = 13$$

$$\|\mathbf{A}\|_\infty = \max\{1 + 2 + 3 + 1, 2 + 3 + 4 + 1, 2 + 5 + 2 + 1, 1 + 1 + 4 + 2\} = 10$$

Problem 2. (10 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$. Find $\kappa_1(\mathbf{A})$ and $\kappa_\infty(\mathbf{A})$.

Solution: It is easy to find

$$\mathbf{A} = \frac{1}{2 \cdot 5 - 3 \cdot 4} \begin{bmatrix} 5 & -4 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} -5/2 & 2 \\ 3/2 & -1 \end{bmatrix}$$

So

$$\kappa_1(A) = \|\mathbf{A}\|_1 \|\mathbf{A}^{-1}\|_1 = 9 \times 4 = 36$$

$$\kappa_\infty(A) = \|\mathbf{A}\|_\infty \|\mathbf{A}^{-1}\|_\infty = 8 \times (9/2) = 36$$